

Week 6 — Hypothesis testing I

Business Forecasting · LIVE deck (slides + exercises in one deck)

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Where we are

C1 Framing → C2 Describing data → **C3 Uncertainty** → C4–C5 Regression → C6 Forecasting

We are in **C3 — uncertainty**. Last week we put a range around an estimate (a CI). This week we **test claims** about a parameter.

- **This week (Testing I): one-sample hypothesis testing** — hypotheses, test statistic, rejection region, Type 1 & 2 errors, power, and p-values.

Fast & hands-on: every test we set up, you run in R (`qnorm`, `pnorm`, `t.test`) — the exercise slides run right here in the browser.

Hypothesis Testing

Hypothesis Testing

Can we get a higher price if cleanliness level is above 4.5?

This is a **hypothesis testing** question: we see which of two competing hypotheses is more likely given the data.

- **Hypothesis** is a claim about a parameter or a distribution.
 1. H_0 **Null Hypothesis**: the claim to be tested, the skeptical perspective.
 - Prices of clean and dirty apartments are equal
 - $H_0 : \mu_c = \mu_d$
 2. H_A **Alternative Hypothesis**: an alternative claim under consideration.
 - Prices of clean apartments are higher
 - $H_A : \mu_c > \mu_d$
- Assume the null is true and see if the data reject it in favor of the alternative.
- You either reject the null, or fail to reject it (never “accept” it).

Hypothesis testing: more examples

Some examples of hypotheses:

- You released a promotion (10% off) to some customers. Do they spend a different amount than those who didn't get it?
- $H_0 : \mu_p = \mu_n$ (promotion group spends the same)
- $H_A : \mu_p \neq \mu_n$ (promotion group spends a different amount)
- A new snack gets an “excessive sugar” sticker if it exceeds 100g of sugar. You take a sample and test.
- $H_0 : \mu_s = 100$ (average sugar is 100 or less)
- $H_A : \mu_s > 100$ (average sugar is more than 100)

Hypothesis testing: why a test at all?

- Suppose we calculated the sample mean for sugar content. It's either below or above 100g.
- Why would we need a fancy test?
- Because we only have a sample, but we want to learn about the population parameter!
- Maybe in our sample it's below 100g, but in the whole population it's above.
- Hypotheses are always about parameters! Never about sample statistics.



Exercise 1 — State the hypotheses

A report claims the true mean monthly electricity bill of Mexican households is **200**. We want to test whether it **differs from** 200. Which pair states the hypotheses correctly?

- $H_0 : \mu = 200$ vs $H_A : \mu \neq 200$
- H_0 : sample mean = 200 vs H_A : sample mean $\neq$ 200
- $H_0 : \mu \neq 200$ vs $H_A : \mu = 200$
- $H_0 : \mu > 200$ vs $H_A : \mu < 200$

$H_0 : \mu = 200$ vs $H_A : \mu \neq 200$.

Hypotheses are always about the population parameter μ , never the sample mean. “Differs from” is two-sided ($\neq$), and the null is the claim being tested ($\mu = 200$).

Testing Procedure

A **test procedure** is a rule based on sample data for deciding whether to reject the null.

It relies on two elements:

- **Test statistic**

- A function of the sample data that summarizes evidence for or against the hypothesis
- We know how it is distributed under the null (e.g. standardized mean)

- **Rejection region**

- Values of the test statistic for which we reject the null
 - These are values that are very unlikely under the null hypothesis
- Typically threshold-based: reject H_0 if the test statistic $> k$

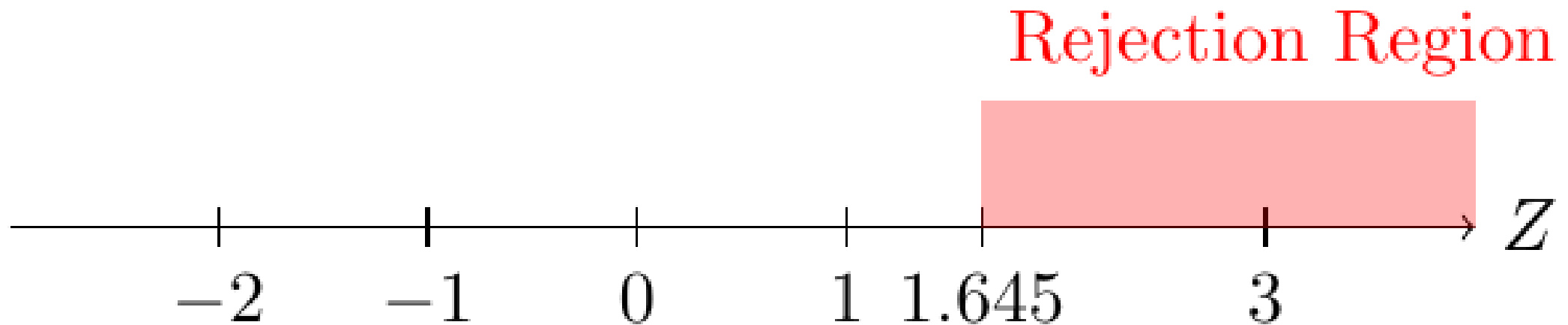
Testing Procedure: Example

Sugar example. Sample $n = 36$, population standard deviation known, $\sigma_x = 20$.

- **Test statistic:**

$$z = \frac{\bar{x} - 100}{\underbrace{\sigma_x / \sqrt{36}}_{SE}}$$

- **Rejection region:** for a 5% significance level, reject if $z \in \{1.645, \infty\}$



The claim we will test

Our running data: `exam_data` — the Fall 2025 **Households & Environment Survey**. One row per household; `electricity_bill_monthly` is the monthly bill in pesos, with some missing values.

A published report claims the true mean monthly bill is **200**.

Before any test: how many valid responses are there, where is the sample centred, and how spread out is it?



Exercise 2 — Describe the sample

Using `exam_data$electricity_bill_monthly`, and dropping the missing values first:

1. How many valid responses are there?
2. What is the sample mean, and the sample standard deviation?
3. What is the standard error of the mean?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Drop the missing values, then the three numbers
```



Exercise 3 — The test statistic by hand

Write the test statistic for $H_0 : \mu = 200$ yourself, from the definition $\frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ — no `t.test()` yet.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Build xbar, s and n from the data, then assemble the statistic
```

Testing Procedure Summary

1. Set up your hypotheses
2. Calculate your test statistic
3. Check it against the rejection region
4. Decision: Reject or fail to reject the null hypothesis

We will now go into the details of each step.

Is that a big number?

The statistic you just computed is a little over 2. Two *what*?

It is only interpretable against a yardstick: **the values the statistic would take if the report were right**. We can build that yardstick by simulating the world in which H_0 is true.



Exercise 4 — Build the null distribution

Simulate 3,000 surveys drawn from a world where the report is right: bills centred at 200, with the same spread as the data, $n = 400$ households each.

1. In each simulated survey, compute the same statistic as in Exercise 3.
2. Draw the histogram of the 3,000 statistics (`freq = FALSE`), and overlay `dnorm` with `curve(dnorm(x), add = TRUE, col = "red", lwd = 2)`.
3. Mark your observed statistic on the picture with `abline(v = ..., lwd = 3)`.
4. Report the mean and standard deviation of the 3,000 statistics, and the share of them at least as far from 0 as yours.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # One simulated survey under H0: sim <- rnorm(400, mean = 200, sd = sd(bill))
2 # Repeat it 3,000 times with replicate(), then plot and measure
```

What you just built

The histogram is the **distribution of the test statistic under H_0** . It came out standard normal — which is why the whole week compares statistics to $N(0, 1)$ quantiles.

Two things follow, and both get names later in this deck:

- The cut-off beyond which we call a value “too extreme” → the **rejection region**.
- The share of the null distribution beyond the observed value → the **p-value**.

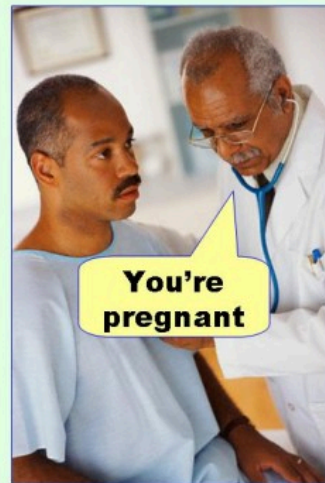
Decision errors

Decision errors

	Do not reject H_0	Reject H_0 in favor of H_A
H_0 true	Okay	Type 1 Error
H_A true	Type 2 Error	Okay

- H_0 : You are not pregnant
- H_1 : You are pregnant

Type I error
(false positive)



Type II error
(false negative)



Decision errors: free shipping

Do people spend more if they get free shipping? (H_0 : they spend the same)

- **Type 1 error**: reject null when H_0 is true
 - Conclude free-shipping customers spend more, while in reality they spend the same
- **Type 2 error**: don't reject null when H_A is true
 - Conclude they spend the same, while free-shipping customers actually spend more
- Trial of a murderer: H_0 innocent, H_1 guilty. Describe error types 1 and 2.
- Type 1: Convict an innocent
- Type 2: Do not convict a murderer
- Probability of a type 1 error is α — the **size** or **significance level**.
 - We don't want to incorrectly reject the null more than $\alpha \cdot 100\%$ of the time.
- Probability of a type 2 error is β — to calculate it, you need the true μ .

Probability of type 1 error

- In the sugar example, suppose the null is true and $\mu_s = 100$.
- You take a sample of 36.
- Say you were unlucky: by chance you got rare high-sugar packages and $\bar{x} = 106$, so $z = \frac{106-100}{20/\sqrt{36}} = 1.8$. You reject the null — an error.
- Q: What is the probability the standardized mean of these 36 is in the rejection region (larger than 1.645)?
- What's the probability of making such an error?

$$\alpha = P_{H_0}(\bar{X} > 105.483) = P_{H_0}\left(\frac{\bar{X} - 100}{20/\sqrt{36}} > \frac{105.483 - 100}{20/\sqrt{36}}\right)$$

$$\stackrel{\text{CLT}}{=} P(Z > 1.645) = 1 - \Phi(1.645)$$

Where exactly does 1.645 come from?

The rejection region for a one-sided test at 5% is a piece of the standard normal — the piece that holds 5% of the area on the right.

Draw it before you see it drawn.



Exercise 5 — Draw the rejection region

1. Draw the standard normal density between -3 and 3 with `curve(dnorm(x), ...)`.
2. Find the cut-off that leaves exactly 5% of the area to its right.
3. Shade that area with `polygon()` and mark the cut-off with `abline(v = ...)`.

Exercise

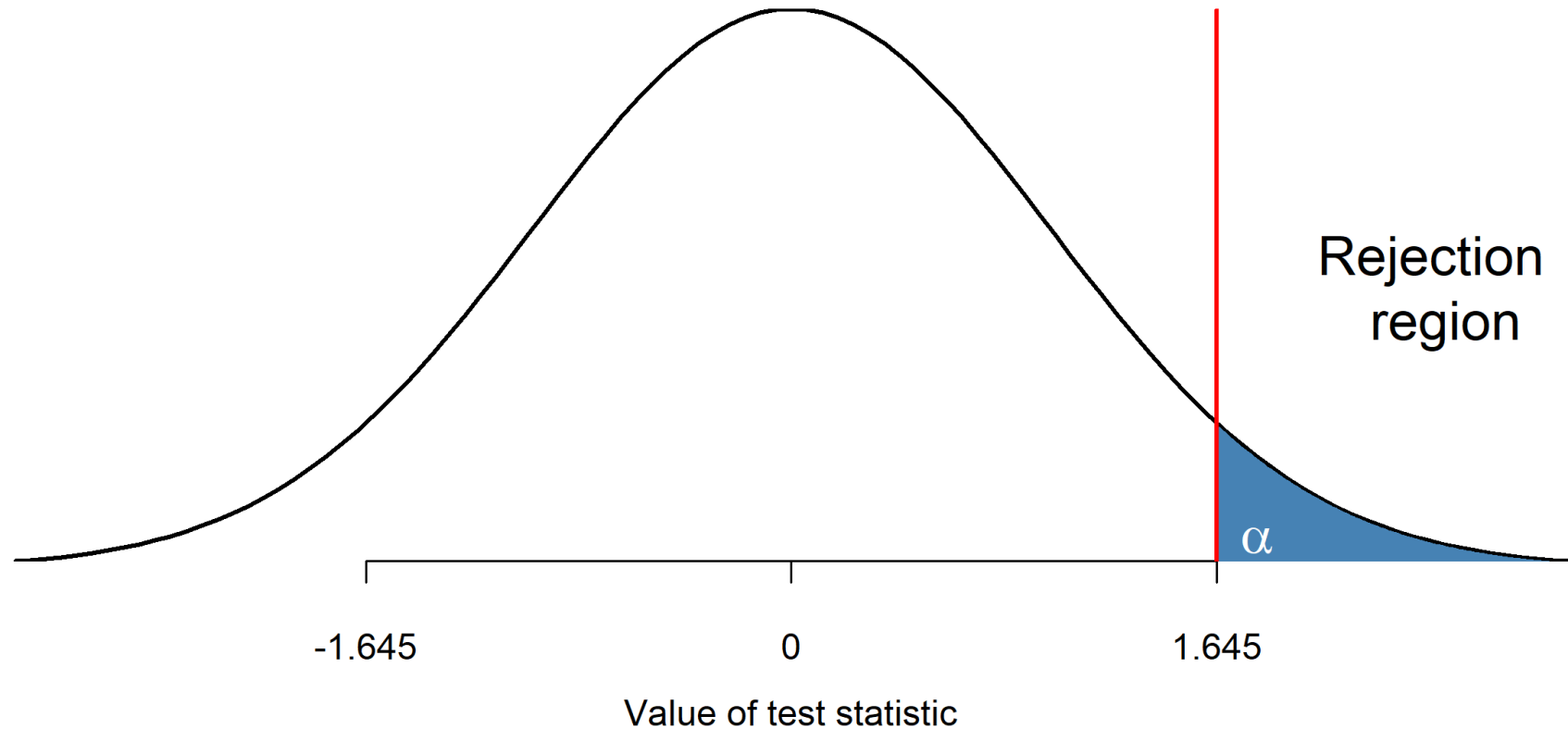
↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # The density:
2 curve(dnorm(x), xlim = c(-3, 3), lwd = 2, xlab = "Test statistic", ylab = "")
3
4 # The cut-off leaving 5% to the right:
5
6 # Shade it:
```

The rejection region and α



- Choosing the critical value determines α (and vice versa). We usually choose α , which fixes the rejection region.
- What is the critical value for $\alpha = 0.1$? For $\alpha = 0.01$? For $\alpha = 0$?



Exercise 6 — Break it: change α

Take the picture you just drew and change **one** thing: the significance level.

1. Compute the one-sided critical value for $\alpha = 0.10, 0.05, 0.01$ and 0.
2. Redraw the density with all three finite cut-offs marked.
3. Report what happens to the **width** of the rejection region as α falls, and what the $\alpha = 0$ case means in practice.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 alphas <- c(0.10, 0.05, 0.01, 0)
2
3 # The critical values:
4
5 # The picture, with the cut-offs marked:
```



Exercise 7 — Type 1 error, made visible

Run 5,000 studies in a world where the null is **true**: sugar content really is $\mu_s = 100, \sigma = 20, n = 36$. Each study computes the statistic and rejects if it exceeds the 5% critical value.

1. What share of these 5,000 honest studies reject a null that is true?
2. Repeat the count at the 10% and the 1% critical values.
3. Draw the histogram of the 5,000 statistics with the 5% cut-off marked.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # One study under a TRUE null:
2 # (mean(rnorm(36, mean = 100, sd = 20)) - 100) / (20 / sqrt(36))
3 # Repeat 5,000 times, then count how many land past each cut-off
```

What the simulation showed

α is not a description of one study. It is the **long-run rate** at which a correct null gets rejected: run the honest study 5,000 times at $\alpha = 0.05$ and roughly 250 of them report a “finding”.

That is the price of the rejection region — and lowering it has a cost of its own.



Exercise 8 — What a Type 1 error means here

In our electricity-bill test ($H_0 : \mu = 200$), what would a **Type 1 error** mean?

- Concluding the mean bill differs from 200 when in truth it equals 200
- Concluding the mean bill equals 200 when in truth it differs
- Collecting a sample that is too small
- Choosing the wrong reference distribution

Concluding the mean bill differs from 200 when in truth it equals 200.

Type 1 error = rejecting a true H_0 (a “false positive”). Its probability is the significance level α . Failing to reject a false H_0 is a Type 2 error.

Probability of type 2 error

- What's the probability our sample mean is below the critical value, if the true sugar content is above 100?
- If I take 10000 samples from a distribution with true mean > 100 , what share of sample means fall below the rejection region?
- In our example:

$$\beta = P(\text{Type 2 error}) = \underbrace{P_{H_A}(z < 1.645)}_{\text{Pr. if alt is true}} = P_{H_A}\left(\frac{\bar{x} - 100}{20/\sqrt{36}} < 1.645\right)$$

- Problem: how is $\frac{\bar{x}-100}{20/\sqrt{36}}$ distributed if the alternative is true?
- CLT does not apply to $\frac{\bar{x}-100}{20/\sqrt{36}}$ because 100 is not the true mean under the alternative.
- We need to specify the true mean. Say $\mu_s = 110$. Then CLT applies to $\frac{\bar{x}-110}{20/\sqrt{36}}$.

Probability of type 2 error (cont.)

- Rewrite so we have $\frac{\bar{x}-110}{20/\sqrt{36}}$ on the left-hand side.

$$\beta = P_{H_A}\left(\frac{\bar{x}-100}{20/\sqrt{36}} < 1.645\right) = P_{H_A}(\bar{X} < 105.483) = P\left(\frac{\bar{X}-110}{20/\sqrt{36}} < \frac{105.483-110}{20/\sqrt{36}}\right)$$

- Now this becomes:

$$P\left(z < \frac{105.483-110}{20/\sqrt{36}}\right) = P(z < -1.3551)$$

- Where z is normally distributed thanks to CLT.

Probability of type 2 error: the general formula

- More generally for this type of test:

$$\beta = P_{H_A}(\bar{X} < \mu_0 + z_{1-\alpha} \frac{\sigma_x}{\sqrt{n}}) = P\left(\frac{\bar{X} - \mu_s}{\sigma_x/\sqrt{n}} < \frac{\mu_0 - \mu_s}{\sigma_x/\sqrt{n}} + z_{1-\alpha}\right)$$

- Depends on the true μ and where the rejection region is.

Trade-off between type 1 and 2 errors

$$\beta = P(\text{Type 2 error}) = P\left(Z < \frac{\mu_0 - \mu_s}{\sigma_x / \sqrt{n}} + z_{1-\alpha}\right) = \underbrace{\Phi\left(\frac{\mu_0 - \mu_s}{\sigma_x / \sqrt{n}} + z_{1-\alpha}\right)}_{\text{Normal CDF}}$$

$$\alpha = P(\text{Type 1 error}) = P(Z > z_{1-\alpha}) = 1 - \underbrace{\Phi(z_{1-\alpha})}_{\text{Normal CDF}}$$

Both formulas contain the same critical value $z_{1-\alpha}$. Moving it moves both numbers — in which directions?



Exercise 9 — Break it: what α costs you in β

Sugar setting: $\mu_0 = 100$, $\sigma = 20$, $n = 36$, and suppose the truth is $\mu_s = 110$.

For $\alpha = 0.10, 0.05$ and 0.01 , compute and tabulate:

1. the critical value $z_{1-\alpha}$,
2. the sample mean at which you would start rejecting, $\mu_0 + z_{1-\alpha}\sigma/\sqrt{n}$,
3. β from the formula on the previous slide,
4. $1 - \beta$.

Then say which way each column moves as α falls.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 mu0 <- 100; mu_true <- 110; sigma <- 20; n <- 36
2 alphas <- c(0.10, 0.05, 0.01)
3
4 # critical value, critical sample mean, beta, and 1 - beta for each alpha
```

The trade-off, stated

If I change the critical value $z_{1-\alpha}$, it has the opposite effect on α and β . Choosing α is choosing which of the two errors you would rather make.

Interactive visualization: <https://shiny.rit.albany.edu/stat/betaprob/>

Power of a test

- Power is the probability of rejecting if the alternative is true.
- Power differs for every possible value of the alternative.
- Power for a one-sided test:

$$Power = P_{H_A}(Reject) = 1 - \underbrace{P_{H_A}(\text{Not Reject})}_{\text{type 2 error}} = 1 - \beta$$

$$Power = 1 - P\left(Z < \frac{\mu_0 - \mu_s}{\sigma_x / \sqrt{n}} + z_{1-\alpha}\right)$$

What makes a test powerful?

Four things could plausibly matter: the sample size n , how far the truth sits from the null, the population spread σ , and the significance level α .

Rather than argue about it — count rejections.



Exercise 10 — Power by simulation

Sugar test again: $H_0 : \mu = 100$ against $\mu > 100, \sigma = 20, \alpha = 0.05$.

1. Write a function that simulates 2,000 studies with a given **true mean** and **n**, and returns the share that reject.
2. Run it for true means 102, 105, 110, 115 with **n = 36**.
3. Run the same four with **n = 100**.
4. Compare your simulated numbers with the formula on the *Power of a test* slide.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 mu0 <- 100; sigma <- 20; zc <- qnorm(0.95)
2
3 # power_sim <- function(mu_true, n, reps = 2000) { ... }
```

What drives power

Power increases if: n increases; the alternative is further from the null; σ decreases; α increases.

The first two are what you just measured; the last one is the trade-off from Exercise 9, seen from the other side.

A/B Test Power: Professional Photos

- Airbnb wants to test if professional photos increase weekly revenue.
- Design:
 - Control: no photos (baseline $\mu_0 = 2000$)
 - Treatment: professional photos (suspected $\mu_a = 2200$)
 - $\sigma = 600, n = 100$ per group, $\alpha = 0.01$
- Steps:
 1. State hypotheses: $H_0 : \mu_T - \mu_C = 0$ vs $H_A : \mu_T - \mu_C > 0$
 2. Find the critical value
 3. Compute power = $1 - \beta$
 4. Find minimum n for 80% power



Exercise 11 — Run the A/B power calculation

Work steps 2–4 in R. The statistic is a **difference of two means**, so its standard error combines both groups:

$$SE = \sqrt{\sigma^2/n + \sigma^2/n}.$$

1. The critical value, expressed as a revenue difference in pesos.
2. The power of the planned test at $\mu_a - \mu_0 = 200$.
3. The n per group that would buy 80% power, from $n = 2\sigma^2 \left(\frac{z_{1-\alpha} + z_{1-\beta}}{\mu_a - \mu_0} \right)^2$.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 mu0 <- 2000; mua <- 2200; sigma <- 600; n <- 100; alpha <- 0.01
2
3 # (1) SE of the difference, then the critical difference
4
5 # (2) power at a true gap of 200
6
7 # (3) n per group for 80% power
```

General Procedure

1. Determine the null and alternative hypotheses
2. Collect your sample of independent observations
3. Choose the appropriate test
4. Calculate the test statistic
5. Compare it to the rejection region
6. Reject or fail to reject the null hypothesis

Let's talk about **appropriate tests**!

Choosing the Appropriate Test

The choice of test statistic and rejection region depends on:

- Type of data we are testing?
 - **Single sample:** $H_0 : \mu = \mu_0$ vs $H_A : \mu \neq \mu_0$
 - **Two independent samples:** $H_0 : \mu_{s1} = \mu_{s2}$ vs $H_A : \mu_{s1} \neq \mu_{s2}$
 - **Paired data:** $H_0 : \mu_{d1} = \mu_{d2}$ vs $H_A : \mu_{d1} \neq \mu_{d2}$
- Can we estimate the standard deviation well?
 - **Yes:** large sample, or known standard deviation
 - **No:** small sample but normal distribution

Hypothesis Testing: Single Sample

Single Sample Test for Mean

General idea: test if the parameter equals/exceeds/falls below some value.

- Ex: $H_0 : \mu = 3$ vs $H_A : \mu \neq 3$
- **Test statistic:** normalized sample mean

$$\text{test statistic} = \frac{\bar{X} - \mu_0}{SE}$$

- SE depends on the estimate of the standard deviation:
 - Standard deviation known: $SE = \frac{\sigma}{\sqrt{n}}$
 - Standard deviation not known: $SE = \frac{s}{\sqrt{n}}$
- **Rejection region:** depends on the distribution
 - Large sample or known variance: $Z_{test} \sim N(0, 1)$ — critical values from standard normal
 - Small sample, normal data: $T_{test} \sim t(n - 1)$ — critical values from Student's t, $n - 1$ df

Single Sample - Mean - Two Sided Test

Hypothesis: $H_0 : \mu = \mu_0$ and $H_A : \mu \neq \mu_0$

Rejection Region: for a test of significance level α , reject

- If large sample or known variance from normal

$$\frac{\bar{X} - \mu_0}{SE} < z_{\frac{\alpha}{2}} \quad \text{or} \quad \frac{\bar{X} - \mu_0}{SE} > z_{1-\frac{\alpha}{2}}$$

- If small sample from normal

$$\frac{\bar{X} - \mu_0}{SE} < t_{(n-1), \frac{\alpha}{2}} \quad \text{or} \quad \frac{\bar{X} - \mu_0}{SE} > t_{(n-1), 1-\frac{\alpha}{2}}$$

Where $z_{1-\frac{\alpha}{2}}$ and $t_{(n-1), 1-\frac{\alpha}{2}}$ are $(1 - \frac{\alpha}{2})$ quantiles of the standard normal and Student's t with $n - 1$ df.



Exercise 12 — Draw the two-sided rejection region

Exercise 5 put all 5% on the right. A two-sided test splits it. Redraw the picture for $H_A : \mu \neq \mu_0$ at $\alpha = 0.05$:

1. Both critical values.
2. Both shaded tails.
3. Check that the two shaded areas together really are 5%.

Then, in one line: which of the two cut-offs, 1.645 or the one you just got, is further from zero — and why?

Exercise

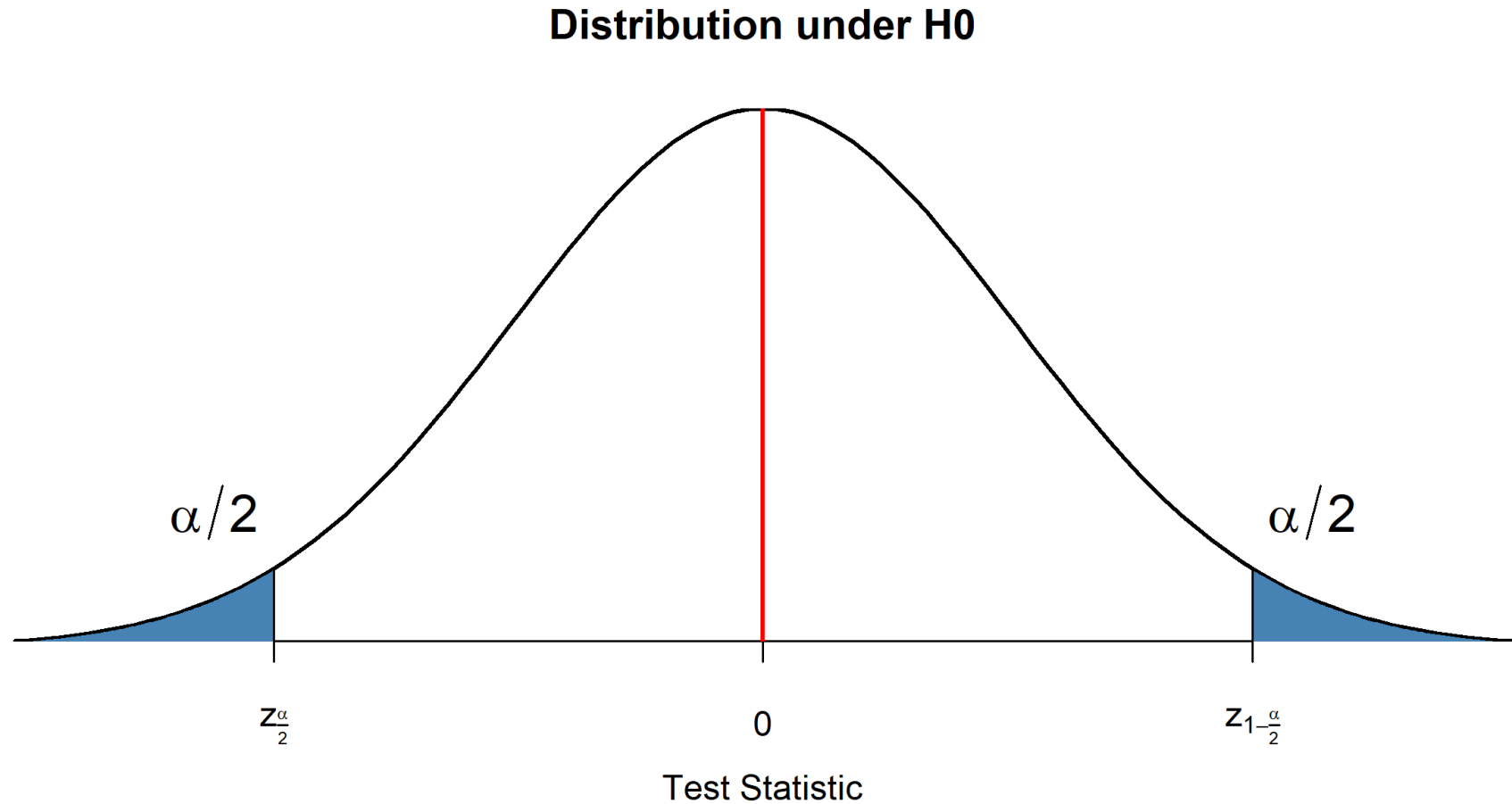
↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # The two cut-offs, then the density with both tails shaded
```

Single Sample - Mean - Two Sided Test: the picture





Exercise 13 — Let R run the test

You built the statistic by hand in Exercise 3 and the cut-offs in Exercise 12. Now do the whole two-sided test in one call: `t.test()` on `exam_data$electricity_bill_monthly` against $\mu_0 = 200$.

Read off the statistic, the degrees of freedom, and compare the statistic with your cut-offs. Does the printed statistic match the one you assembled by hand?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # One call, then compare its statistic with your hand-built one
```

Two answers to the same question

Last week you built a 95% confidence interval for a mean. This week you test whether the mean equals 200. Two different procedures, same data.

Can they contradict each other?



Exercise 14 — Test and confidence interval

1. Pull the 95% confidence interval out of the `t.test()` output and check whether 200 lies inside it.
2. Compare that verdict with the test's own decision at $\alpha = 0.05$.
3. **Break it:** rerun with `conf.level = 0.99` (i.e. $\alpha = 0.01$). Does the interval still exclude 200? Does the decision still agree?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # The 95% interval and the 5% decision
2
3 # Now the 99% interval and the 1% decision
```

Link between Tests and Confidence Intervals

Test for μ with $H_0 : \mu = 4$ and $H_A : \mu \neq 4$.

- Population standard deviation known, $\sigma = 3$. Sample of 49 with mean $\bar{x} = 3$.
- Test statistic: $z_{test} = \frac{3-4}{\sigma/\sqrt{49}} = \frac{3-4}{3/7} = -2.34$
- Critical values at $\alpha = 0.05$ are -1.96 and 1.96 , hence we reject.
- The 95% confidence interval for the mean:

$$\left\{3 - 1.96\frac{\sigma}{\sqrt{49}}, 3 + 1.96\frac{\sigma}{\sqrt{49}}\right\} = \{2.16, 3.84\}$$

- It does not contain the null hypothesis.

Link between two sided Tests and Confidence Intervals

- More generally: the $1 - \alpha$ CI doesn't contain the null $\iff$ the null would be rejected with a test of significance α .

https://kzaremba.shinyapps.io/Hypothesis_Confidence/

- Mathematically: let μ_0 be the null. In a two-sided test we reject at α if

$$\left| \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \right| > z_{1-\frac{\alpha}{2}}$$

- Which can be rewritten as:

$$\mu_0 < \bar{X} - z_{1-\frac{\alpha}{2}} \frac{s}{\sqrt{n}} \quad \text{or} \quad \mu_0 > \bar{X} + z_{1-\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$$

Single Sample - Mean - One sided - Case 1

Hypothesis: $H_0 : \mu = \mu_0$ and $H_A : \mu < \mu_0$

- This includes any null like $\mu \geq \mu_0$. Why?
- If you rejected μ_0 at α , you would reject for sure anything larger than μ_0 .

Rejection Region: for a test of significance level α , reject

- If large sample or known variance from normal

$$\frac{\bar{X} - \mu_0}{SE} < z_\alpha$$

- If small sample from normal

$$\frac{\bar{X} - \mu_0}{SE} < t_{(n-1),\alpha}$$

Where z_α and $t_{(n-1),\alpha}$ are α quantiles of the standard normal and Student's t with $n - 1$ df.

Move all of α into one tail

The two-sided picture splits α into two halves. A one-sided alternative puts all of it on one side.

Take your two-sided code and change only where the shading goes.



Exercise 15 — Break it: one tail instead of two

Starting from your Exercise 12 code, at $\alpha = 0.05$:

1. Draw the **left-tail** picture ($H_A : \mu < \mu_0$) and give its critical value.
2. Draw the **right-tail** picture ($H_A : \mu > \mu_0$) and give its critical value.
3. Put the three critical values side by side (two-sided, left, right) and say which alternative makes a statistic of -1.8 significant, and which does not.

Exercise

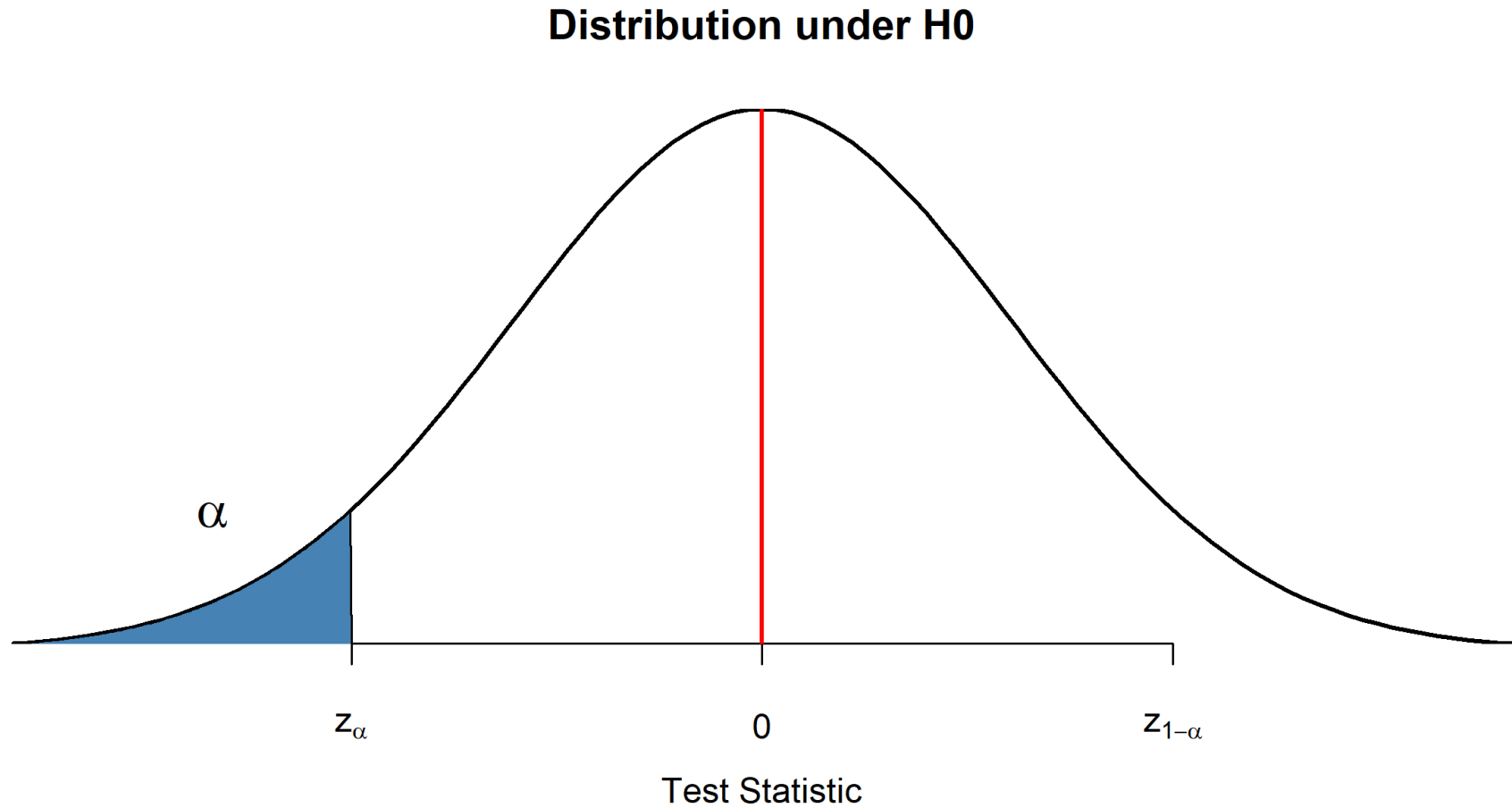
↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # par(mfrow = c(1, 2)) puts the two pictures next to each other
```

Single Sample - Mean - One sided - Case 1: the picture



Single Sample - Mean - One sided - Case 2

Hypothesis: $H_0 : \mu = \mu_0$ and $H_A : \mu > \mu_0$

- This includes any null like $\mu \leq \mu_0$. Why?
- If you rejected μ_0 at α , you would reject for sure anything smaller than μ_0 .

Rejection Region: for a test of significance level α , reject

- If large sample or known variance from normal

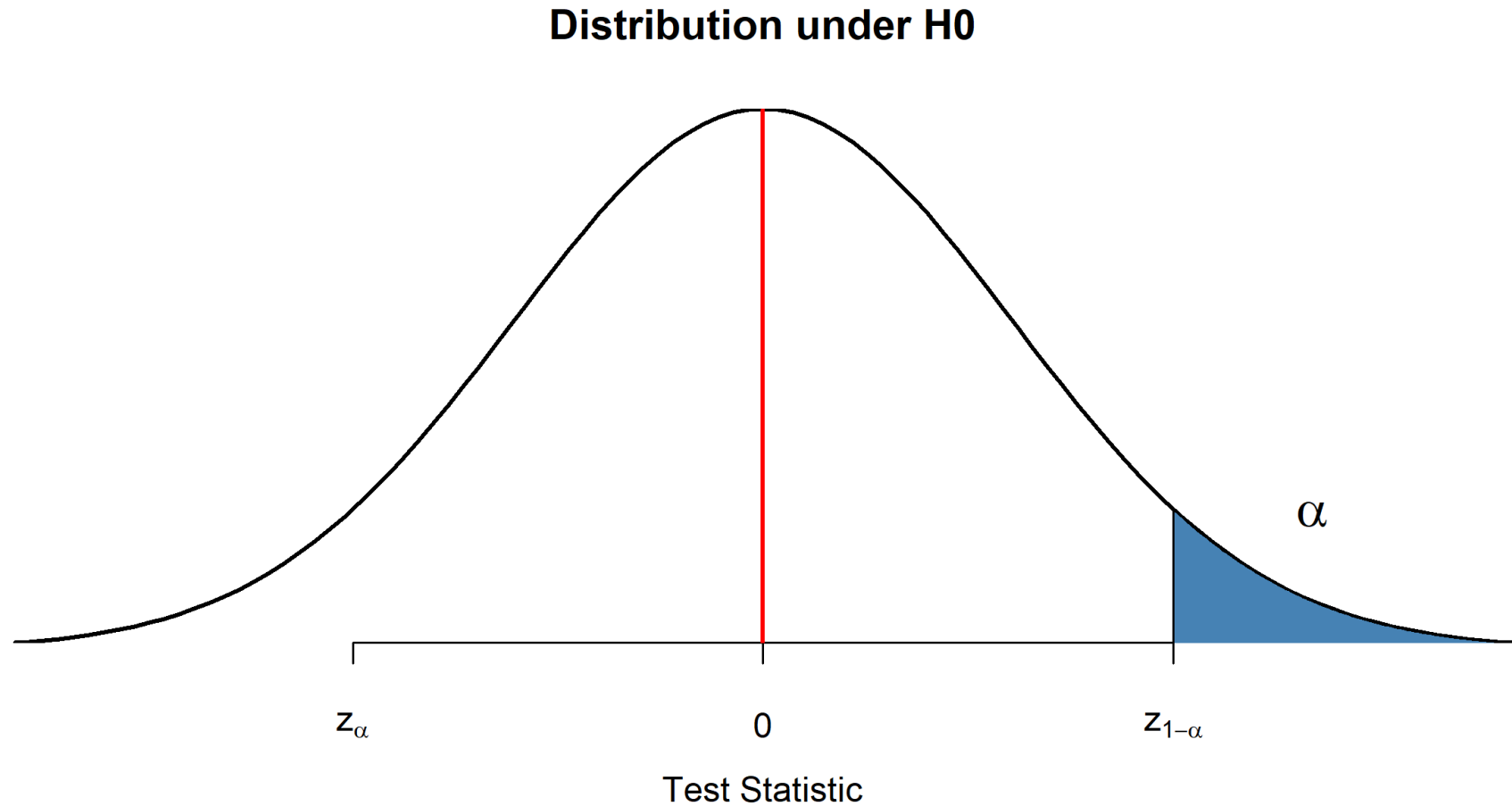
$$\frac{\bar{X} - \mu_0}{SE} > z_{1-\alpha}$$

- If small sample from normal

$$\frac{\bar{X} - \mu_0}{SE} > t_{(n-1), 1-\alpha}$$

Where $z_{1-\alpha}$ and $t_{(n-1), 1-\alpha}$ are $(1 - \alpha)$ quantiles of the standard normal and Student's t with $n - 1$ df.

Single Sample - Mean - One sided - Case 2: the picture



P-values

P-values

Definition: the p-value is the probability, under the null hypothesis, of obtaining a test statistic at least as extreme as the one computed from the sample.

Depends on: your hypothesis, and the calculated test statistic.

- Suppose under H_0 the test statistic is $Z_{test} \sim N(0, 1)$.
- One-sided tests:
 - $H_A : \mu > \mu_0$: $p\text{-value} = P(Z_{Test} \geq \text{computed statistic})$
 - $H_A : \mu < \mu_0$: $p\text{-value} = P(Z_{Test} \leq \text{computed statistic})$
- Two-sided test:
 - $H_A : \mu \neq \mu_0$: $p\text{-value} = 2P(Z_{Test} \geq |\text{computed statistic}|)$

P-values: Example

- Test $H_0 : \mu = 5$ vs $H_A : \mu > 5$.
- In a sample of 36 we have $\bar{x} = 5.5$ and $s = 2$.
- The test statistic is distributed normally:

$$p\text{-value} = P(Z > \frac{5.5 - 5}{2/\sqrt{6}}) = P(Z > 1.5) = 0.066$$

Intuitively: the probability that a statistic we calculated (or a more extreme value) could arise just by chance if the null is true.

P-value in two-sided test

- In a two-sided test, the p-value accounts for **both tails**.
- We double the one-tail probability: $p\text{-value} = 2P(Z \geq |z_{test}|)$.

Interactive visualization: <https://rpsychologist.com/pvalue/>

Exercise 16 — The p-value from the same data

(a) Compute the **two-sided** p-value for the electricity-bill test three ways: from your hand-built statistic with `pnorm()`, from the same statistic with `pt()`, and from `t.test()`. Do all three land in the same place, and why would `pnorm` and `pt` differ at all?

Exercise

 Start Over

 Show Hint

 Run Code

```
1 # Rebuild the statistic, then the three routes to a two-sided p-value
```

(b) Now the **one-sided** p-values from that same statistic: for $H_A : \mu > 200$ and for $H_A : \mu < 200$. Confirm each with the matching `alternative =` argument of `t.test()`, and state how the three p-values relate to each other.

Exercise

 Start Over

 Show Hint

 Run Code

```
1 # The two one-sided p-values, by hand and via t.test()
```

P-values and Test Significance

- In our example we calculated the p-value of 0.066.
- Would a test at $\alpha = 0.05$ reject the null?
- Any test with significance level $\alpha > p\text{-value}$ would reject the null.

Another way to think about p-values:

- The p-value is the smallest significance level α at which H_0 can be rejected.

(3) The average grade among fourth graders on a math aptitude test in a certain state is 62.5 out of 100 points. (Assume 62.5 is the population average.) An educational company has developed an app for elementary-school math and would like to test whether the app improves average performance on the aptitude test among fourth graders. Let μ_X denote the population mean of test scores among students who use the app.

- To provide statistical evidence that the app *increases* the true average of test scores, what are the appropriate one-sided null hypothesis and alternative hypothesis?
- The company provides the app to 15 randomly selected students. The sample mean and sample standard deviation of test scores are 68.3 and 13.5, respectively. Under the assumption that the test scores are i.i.d. and normally distributed, would the t -test of the null hypothesis in (a) be rejected at the 5% level? at the 10% level?
- What is the p-value associated with the t -test in (b)?



Exercise 17 — Decision at $\alpha = 0.05$

Your two-sided test on the electricity bills returned the p-value you computed in Exercise 16. Testing at $\alpha = 0.05$, what do you conclude?

- Reject H_0 — at $\alpha = 0.05$ there is evidence the mean bill differs from 200
- Fail to reject H_0 — p is above 0.05
- Accept H_0 — the mean equals 200 exactly
- Nothing can be decided without knowing the sample size

Reject H_0 — at $\alpha = 0.05$ there is evidence the mean bill differs from 200.

Rule: reject H_0 when p-value $< \alpha$. Here $0.044 < 0.05$, so we reject — though it is a close call, and at $\alpha = 0.01$ the same data does not reject (Exercise 14). We never “accept” H_0 ; we only reject or fail to reject.

Exam spotlight — Fall 2025

Midterm 1

Real Midterm 1 questions on the **Households & Environment Survey** ($n = 13,255$). We solve these in class on `Data/households_environment.rda` (object `exam_data`).

→ [More exam practice at the end of this deck](#)

Exam question — Q9 & Q10

Q9. A report claims the true mean monthly electricity bill is 200. Your sample mean is 206. Why can't we immediately conclude the report is wrong?

Q10. State H_0 and H_A for testing whether the true mean monthly electricity bill **differs from** 200.

A sample mean of 206 could arise by chance even if $\mu = 200$ — sampling variability. We need a **test** (Q10: $H_0 : \mu = 200$ vs $H_A : \mu \neq 200$).

Exam question — Q12

Q12. Using the electricity-bill data ($n = 13,255$), **compute the test statistic** for $H_0 : \mu = 200$ vs $H_A : \mu \neq 200$, and give your decision.

Do it here, exam-style: statistic first, then the decision at **two** significance levels.

Exercise

 Start Over

 Show Hint

 Run Code

1 # The statistic, the two critical values, and the two decisions

Practice now: the one-sample z/t problems in the Week 6 (Testing I) set at the end of this deck.

Exercise 18 — The whole week, one question

You advise the energy regulator. The published report says the average Mexican household pays **200 pesos** a month for electricity. Your director asks:

“Is that number defensible, or should we correct it?”

Answer with a **number**, a **picture**, and **one sentence**.

Exercise

 Start Over

 Show Hint

 Run Code

```
1 # 1. A number: the evidence against 200 -- statistic, p-value, and an interval.
2
3 # 2. A picture: where the observed statistic falls in the distribution H0 predicts.
4
5 # 3. One sentence: what do you write in the memo?
```

What you must be able to do this week

- **State H_0 and H_A** about a parameter (never a sample statistic), one- or two-sided.
- **Build the test statistic** $\frac{\bar{X} - \mu_0}{SE}$ and pick the right reference distribution (normal vs Student's t).
- Define the **rejection region** for a chosen α and reach a **decision**.
- Explain **Type 1 vs Type 2 errors**, the significance level α, β , and **power** — and their trade-off.
- Compute and interpret a **p-value**, and use the **test $\Leftrightarrow$ confidence interval** equivalence.

Next week: comparing **two** groups — difference in means, paired data, correlation, and A/B testing.

Appendix — optional (extra detail)

Appendix: from the statistic back to the sample mean

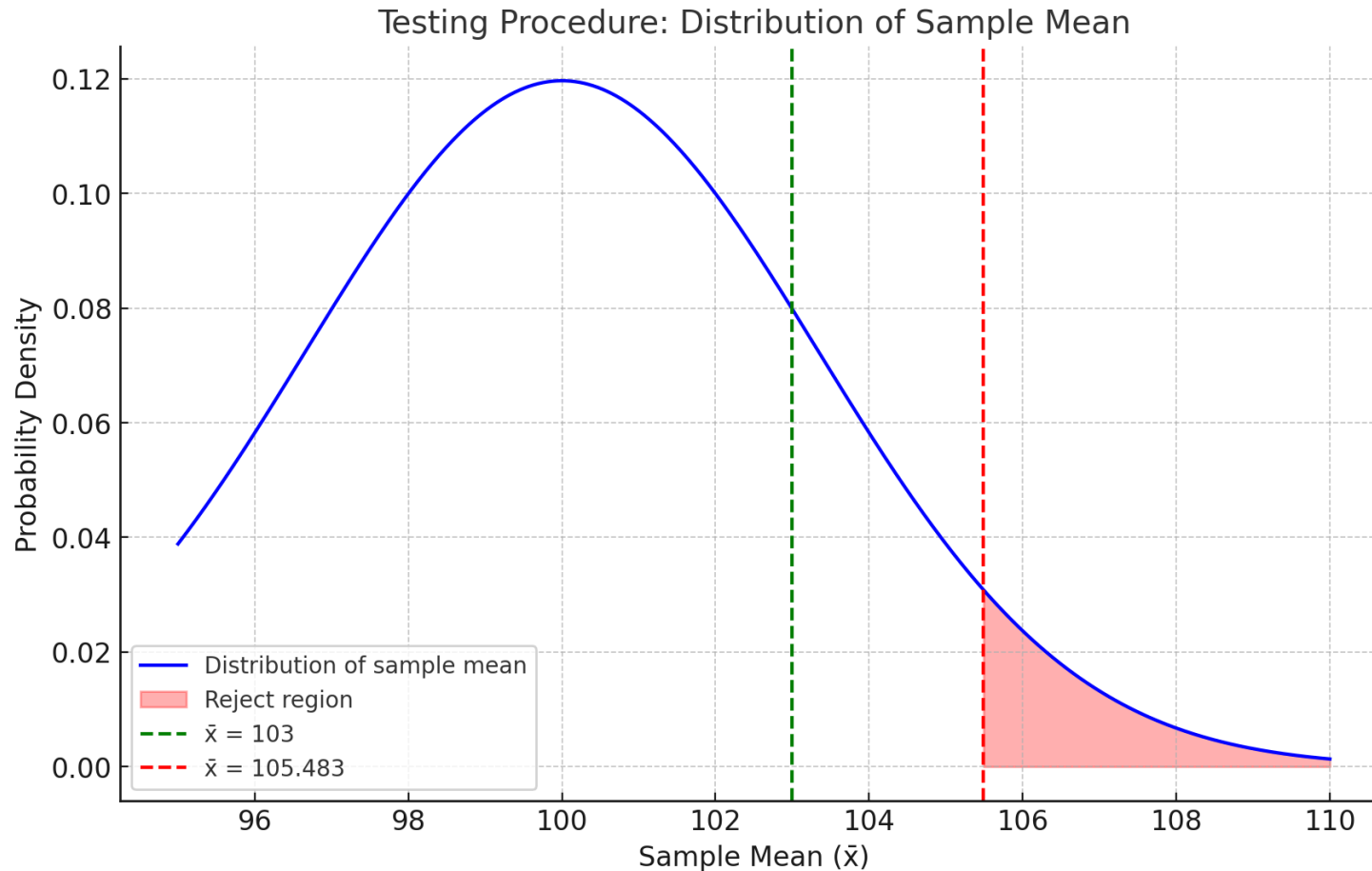
- Suppose I reject if $z > 1.645$, with $\sigma_x = 20$. How large must the mean be to reject?

Reject if:

- $\frac{\bar{x} - 100}{20/\sqrt{36}} > 1.645$
- Or equivalently if $\bar{x} > 105.483$
- Those two formulations are equivalent.
- Why don't we reject even if $\bar{x} = 103$?

Appendix: why 103 is not enough

- Because there is randomness in the sample:
- Even if $\mu_s = 100$, we could get a sample with $\bar{x} > 100$.



- Why $\bar{x} > 105.483$ though? Formally later, but it's very unlikely to get a sample mean that large if $\mu_s = 100$.

Exam practice

These problems mirror the Week 6 exercise set and the exam — set up the test, compute in R, decide; the **Solution** button reveals the worked solution.

[← Back to the exam spotlight](#)

Exam practice 1 — Predicted income: t-statistic and p-value

A random sample of 15 Mexican university students reports predicted first-job income: $\bar{x} = 46,000$ pesos, $s = 9,000$ pesos. [14]

(a) Compute the t -statistic for $H_0 : \mu_X = 48,000$. (c) Compute the p-value (two-sided).

Exercise ↺ Start Over 💡 Show Hint ▶ Run Code

```
1 xbar <- 46000; s <- 9000; n <- 15; mu0 <- 48000
2 # (a) the standardized distance between the sample mean and the null value
3
4 # (c) the two-sided p-value from the right reference distribution
```

(b) Do you reject H_0 at 5% in a two-sided test?

No — fail to reject. The critical value is $t_{0.025, 14} = 2.145$ (`qt(0.975, 14)`), and $|t| = 0.861 < 2.145$. Equivalently, the p-value $0.404 > 0.05$.

Exam practice 2 — Predicted income: a larger sample

Same setting as the previous slide ($\bar{x} = 46,000$, $s = 9,000$, $H_0 : \mu_X = 48,000$). [14]

(d) If $n = 1,000$ instead of $n = 15$, would you reject at 5%?

Exercise

 Start Over

 Show Hint

 Run Code

```
1 xbar <- 46000; s <- 9000; mu0 <- 48000
2 # Redo Exam practice 1 with n = 1000 and compare the two verdicts
```

(e) For (d), is the normality assumption necessary?

No. With $n = 1,000$ the Central Limit Theorem makes the sampling distribution of $\bar{X}$ approximately normal regardless of the underlying distribution of income. (With $n = 15$ we needed normality to justify the t distribution.)



Exam practice 3 — Candy taste test

250 ITAM students take a blind taste test of green vs. red candies: **140 prefer green**, 110 prefer red. [15]

(a) Compute the asymptotic 95% CI for π . (c) Compute the p-value for the test of equal preference.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 n <- 250; green <- 140
2 # (a) the sample proportion, its standard error, and the 95% interval
3
4 # (c) the test statistic for equal preference, then its two-sided p-value
```

(b) State the null hypothesis for equal preference.

$H_0 : \pi = 0.5$ — students are equally likely to prefer green or red. The alternative is two-sided, $H_A : \pi \neq 0.5$.



Exam practice 4 — Smoking rate

The smoking rate among Mexican adults aged 25–44 is **10.5%**. A public health researcher shows an informational video to a random sample. [16]

(a) State the null and alternative hypotheses for testing whether the video **decreases** smoking prevalence.

$H_0 : \pi \geq 0.105$ vs $H_1 : \pi < 0.105$ — a one-sided (lower-tail) test: the skeptical position is that the video does not reduce smoking.

(b) If the follow-up survey finds **8.5%** smokers, how large must the sample be to reject the null at 5%?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 pi0 <- 0.105; pihat <- 0.085
```

```
2 # The rejection condition is an inequality in n -- rearrange it and solve for n
```



Exam practice 5 — Gubernatorial election

In a gubernatorial election in Mexico, older voters (65+) support candidate A with probability 68%; younger voters (≤ 65) support candidate B with probability 55%. Suppose 400 older and 1,100 younger voters turn out, A and B are the only options, and everyone votes. [17]

(a) What is the asymptotic distribution of the sample proportion of votes for candidate A? (b) What is the approximate 95% probability interval for A's sample vote share? (c) What is the approximate probability that candidate A wins?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 n_old <- 400; n_young <- 1100
2 # (a) the overall probability of an A vote, and the SE of A's vote share
3
4 # (b) an interval around it holding about 95%
5
6 # (c) the chance A's share lands above one half
```